Huawei Access Node Optical Transport Failure Troubleshooting SOP (BAS-ACC-07)

Node: BAS-ACC-07 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_7777

Description

The BAS-ACC-07 node is experiencing an Optical Transport Failure, as indicated by the ALARM_7777 code, which signifies a critical severity level. This failure domain occurs on the Huawei Access Node due to the inherent protocol and hardware mechanics of the Optical Transport Network (OTN). Specifically, the failure is likely caused by the degradation of the SFP/XFP transceivers, fiber cuts, or Rx/Tx power loss within the OTN. The Huawei Access Node is a critical component of the network infrastructure, responsible for providing access to the network for end-users. The OTN protocol relies on the proper functioning of the SFP/XFP transceivers to transport optical signals between nodes. If these transceivers degrade or are damaged, the OTN protocol cannot function correctly, leading to the Optical Transport Failure. Additionally, fiber cuts or Rx/Tx power loss can also cause the OTN protocol to fail, leading to the same failure domain. It is essential to identify and rectify the root cause of the failure to ensure the network remains operational and reliable.

Pre-requisite Checks

- check: Verify that the redundant link is not affected by the failure.

- check: Ensure that the remote hands are dispatched to the Basavanagudi POP to assist in the troubleshooting and repair process.

- check: Check the environmental conditions at the Basavanagudi POP to ensure that they are within the recommended operating range for the Huawei Access Node.

Potential Root Causes

Cause	Probability
Degradation of SFP/XFP transceivers	High
Fiber cuts	Medium
Rx/Tx power loss within the OTN	Low

Diagnostic Commands & Expected Outputs

```
Command: display opt trans status, Expected Output: Transceiver status: Normal, Description: This command displays the status of the SFP/XFP transceivers, which can help identify if the transceivers are degraded or damaged.

Command: display opt trans power, Expected Output: Rx power: -30 dBm, Tx power: 0 dBm, Description: This command displays the power levels of the Rx and Tx signals, which can help identify if there are any issues with the OTN protocol.
```

```
Command: display opt trans fiber status, Expected Output: Fiber status: Normal, Description: This command displays the status of the fiber, which can help identify if there are any fiber cuts or issues with the fiber.
```

Resolution Steps

Step 1: Step: Replace the degraded SFP/XFP transceivers, Command: replace opt transceiver 1/1/1, Description: Replace the degraded SFP/XFP transceivers with new ones to ensure that the OTN protocol can function correctly.

Step 2: Step: Repair the fiber cuts, Command: repair opt fiber 1/1/1, Description: Repair the fiber cuts to ensure that the OTN protocol can function correctly.

Step 3: Step: Adjust the Rx/Tx power levels, Command: adjust opt trans power 1/1/1, Description: Adjust the Rx/Tx power levels to ensure that the OTN protocol can function correctly.

Post-Check Verification

```
Command: display opt trans status, Expected Output: Transceiver status: Normal, Description: Verify that the SFP/XFP transceivers are functioning correctly.

Command: display opt trans power, Expected Output: Rx power: -20 dBm, Tx power: 0 dBm, Description: Verify that the power levels of the Rx and Tx signals are within the recommended range.

Command: display opt trans fiber status, Expected Output: Fiber status: Normal, Description: Verify that the fiber is functioning correctly.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP transceivers, Command: restore opt transceiver 1/1/1, Description: Restore the original SFP/XFP transceivers to their previous state in case the replacement did not resolve the issue.

- Step: Repair the original fiber cuts, Command: repair opt fiber 1/1/1, Description: Repair the original fiber cuts to their previous state in case the repair did not resolve the issue.

- Step: Reset the Rx/Tx power levels, Command: reset opt trans power 1/1/1, Description: Reset the Rx/Tx power levels to their previous state in case the adjustment did not resolve the issue.